

Pulse 4K

Ref. PLS-4K



ANALOG WAY®

4K60 multi-layer video mixer and seamless presentation switcher, with ten inputs, two PGM outputs and one dedicated Multiviewer



TAA Compliance: This product is manufactured in France, a TAA-designated country, and is fully compliant with the Trade Agreements Act (17 U.S.C. & 2501-2581).



Pulse 4K at a glance

- ▶ Compact 2RU rugged chassis
- ▶ Ten inputs (8x 4K60 + 2x 1080p inputs) + two 4K60 multiplug outputs
- ▶ 2 modes : Matrix (2 PGM) or Switcher (1 PGM + 1 AUX)
- ▶ Dedicated multiviewer with resizable widgets
- ▶ Real-time 10-bit 4:4:4 video processing compliant with HDR10 & HLG
- ▶ Live Background + up to four 4K split layers (or two 4K mixing layers)
- ▶ RTMP-based livestreaming of any input or output up to 1080p30
- ▶ Intuitive HTML5-based user interface with live source thumbnails
- ▶ Advanced audio processing and dynamic routing
- ▶ Optional analog and Dante™ audio networking card (32x32 channels)

**10
Inputs**

**2 PGM
outputs**

**4K60
4:4:4**

**1
Multi-
viewer**

**HDR
3D LUT**

**IP
Stream**

**Live
back-
ground**

**up to
4 layers
4K**

**HDMI™
2.0**

DP 1.2

12G-SDI

Midra™ 4K models	QuickVu 4K	QuickMatrix 4K	Pulse 4K	Eikos 4K
Inputs	10	10	10	10
Mirrored outputs (HDMI + SDI)	2	2	2	2
Dedicated multiviewer	✓	✓	✓	✓
Mixer mode (1x PGM + 1x AUX)	✓		✓	✓
Matrix mode (2x PGM)		✓	✓	✓
Hard/Soft Edge mode (1x PGM)				✓
Background layer	image	image	live source	live source
4K split layers per system	2	4	4	4
4K foreground layer (still image)	✓	✓	✓	✓
IP RTMP Streaming	✓	✓	✓	✓

Versatile 4K Connectivity

The **Pulse 4K** is a powerful 4K60 multi-layer mixer and seamless presentation switcher equipped with ten inputs, including eight 4K60 inputs and two 1080p inputs with user-selectable HDMI™ and 3G-SDI connectors. The **Pulse 4K** has two 4K60 mirrored HDMI and SDI outputs that can be configured either as Program & AUX or as two independent Program outputs. In addition, the **Pulse 4K** has a dedicated multiviewer to monitor your Program and Preview screens as well as all your inputs.

Quality without Compromise

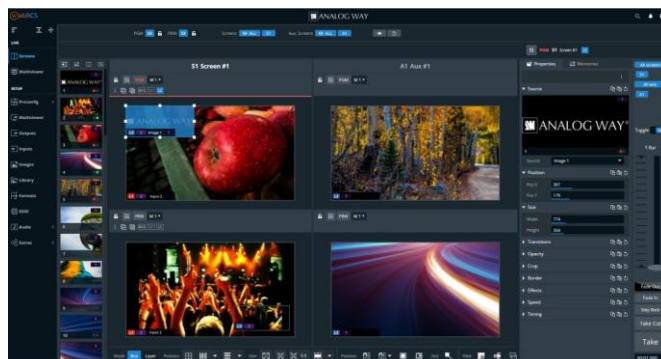
Based on the extremely robust and reliable **Midra™ 4K** (FPGA-based hardware) platform as well as on the 5th generation scaling engine inherited from the Aquilon series, the **Pulse 4K** features ultra-low latency 4K60 10-bit 4:4:4 image processing, uncompromising video quality, true seamless switching, HDR support and HDCP 2.2 compatibility, which makes it an ideal solution for your medium-sized live events and fixed in-installations.

Smart Functionalities

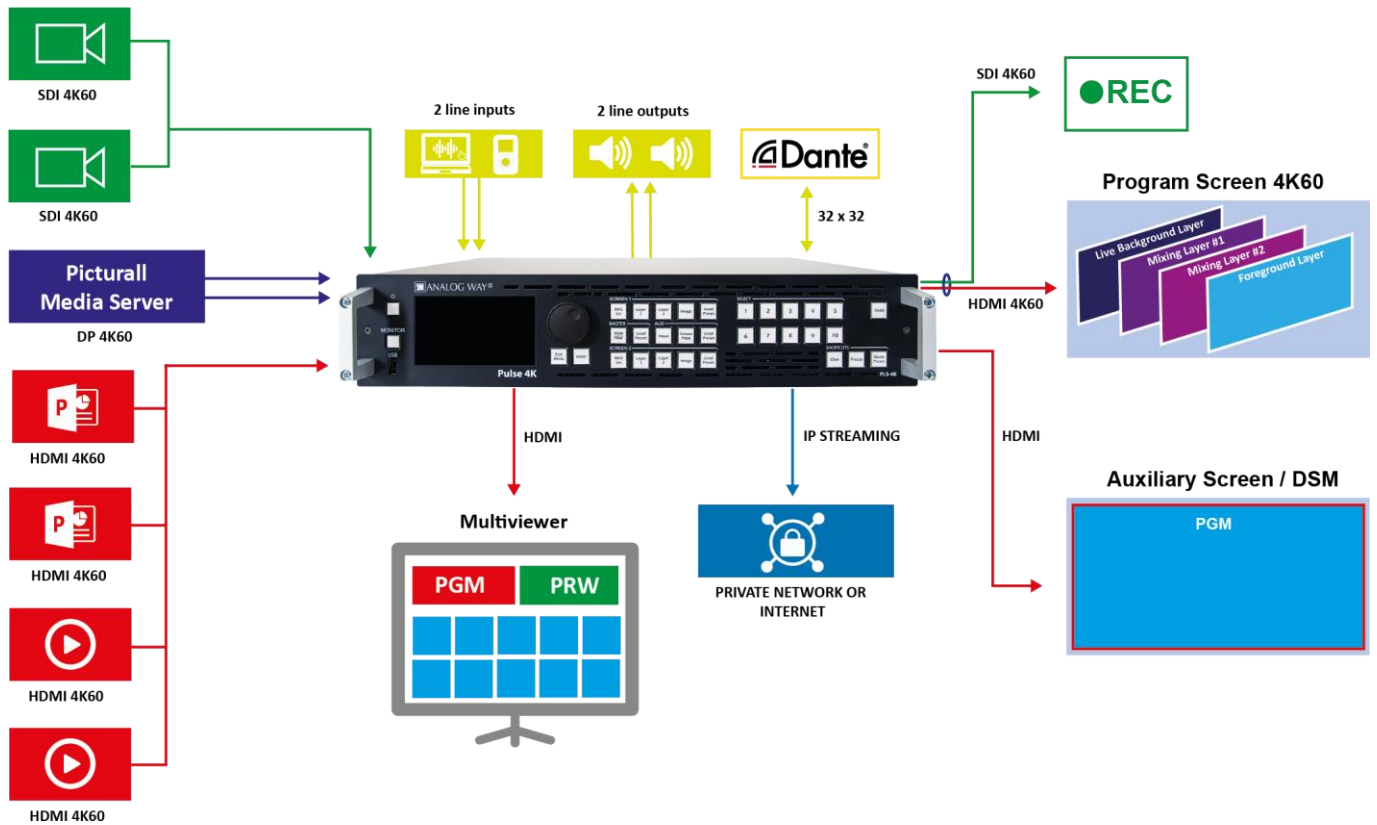
The **Pulse 4K** offer dozens of powerful functionalities that will help you to produce elegant AV presentations: custom output formats, livestreaming of any Input or Output, Luma/Chroma keying, Cut and Fill at input level, AOI, HFR up to 144Hz... The **Pulse 4K** also allows to de-embed or embed digital audio on all the inputs and outputs. An audio option is available to add Dante™ audio networking support to the **Pulse 4K**.

Powerful and Flexible Control Options

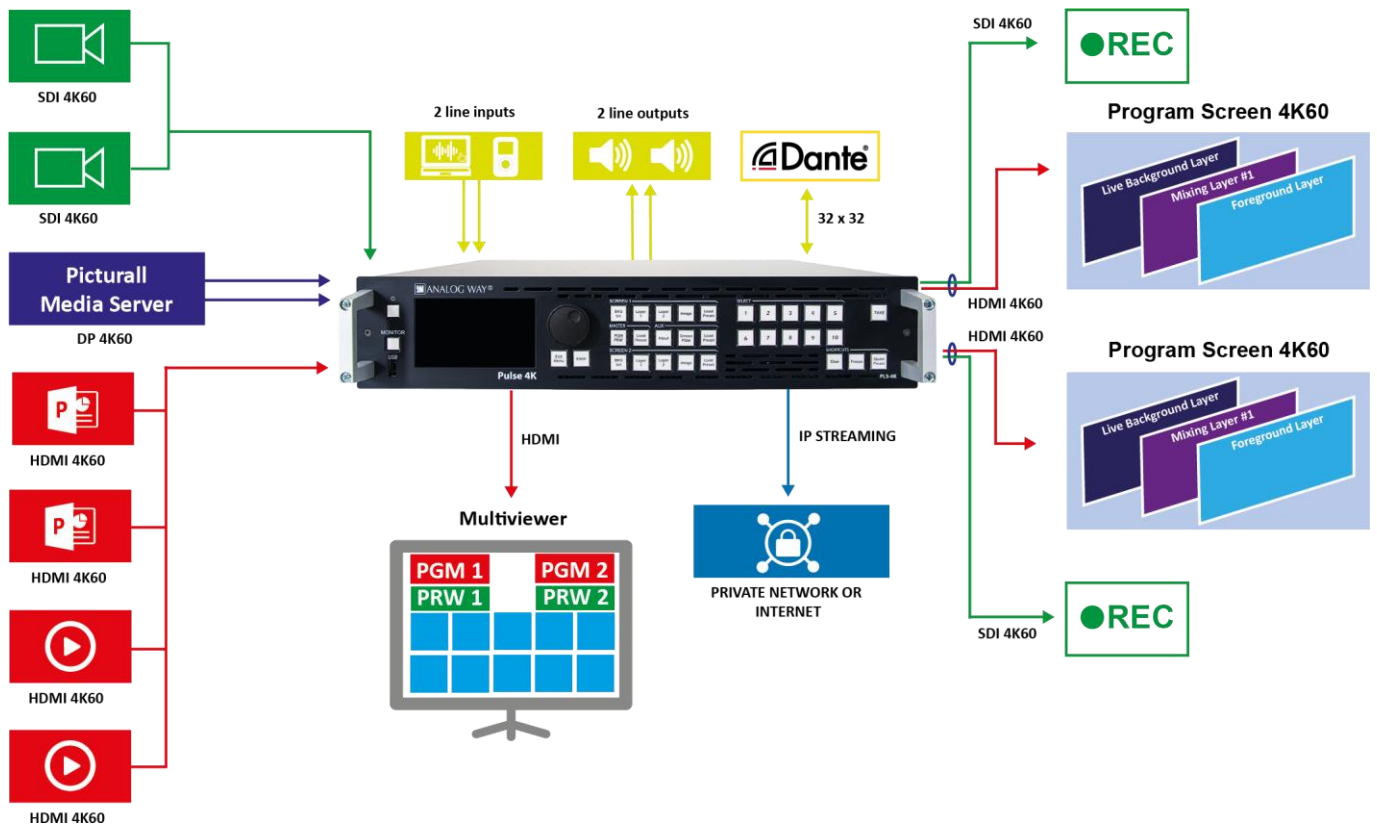
The **Pulse 4K** has been designed to be entirely operated from the easy-to-use front panel. You can also use the on-board intuitive HTML5-based user interface, the Web RCS, designed to allow for the ease of setup and to ensure flawless control of your presentations. The **Pulse 4K** also features Ethernet control via a standard TCP/IP socket connection supported by all major third-party control systems as well as SNMP management. The **Pulse 4K** can also be controlled by the free Crestron driver as well as by AW VideoCompositor, a unique drag and drop customizable Crestron user interface. Additionally, the **Pulse 4K** can be operated by a comprehensive range of remote control panel options such as the powerful event controller RC400T or the compact Control Box³ and Shot Box². The **Pulse 4K** is also compatible with various control software or controller such as Companion or SKAARHOJ Air Fly Pro.



Pulse 4K - Mixer mode with Dante audio option



Pulse 4K - Matrix mode with Dante audio option



The Pulse 4K has an extensive ecosystem for control and management



Optional Analog and Dante™ audio networking card

- ▶ De-embed audio from sources and route directly to Dante™ network
- ▶ Re-embed audio from external audio processor for sending to display
- ▶ 32x32 Dante™ channels @48 kHz
- ▶ Dual redundancy Ethernet ports
- ▶ 2 analog stereo mini jack line inputs 1/8" (3.5 mm)
- ▶ 2 analog stereo mini jack line outputs 1/8" (3.5 mm)



Key features

10 inputs including 8x 4K60 inputs and 2x 2K60 inputs

Two 4K60 PGM outputs with mirrored 12G-SDI and HDMI 2.0 plugs

Mixer mode:

- ▶ 1 PGM output up to 4K60, with 1 live background + 2 mixing 4K layers (or 4 split layers) + 1 foreground 4K image layer
- ▶ 1 AUX output up to 1080p60

Matrix mode:

- ▶ 2 PGM outputs up to 4K60, with 1 live background + 1 mixing 4K layer (or 2 split layers) + 1 foreground 4K image layer per output

1 dedicated multiviewer with resizable widgets, including clocks, countdown and timers (max. resolution 2560x1600@60)

4K60 10-bit 4:4:4 processing - ultra low latency (16 msec @60Hz)

Compliant with HDR10 and HLG - Built-in SDR/HDR converters

Colorspace conversion & color corrections based on 3D LUTs

Seamless crossfade on all live sources

RTMP-based livestreaming of any input or output up to 1080p30

Versatile 4K connectivity: HDMI 2.0, DP1.2, 12G-SDI

8x 4K concurrent still images with alpha channel support

Advanced EDID management

Compatible with HDCP 1.4 and HDCP 2.2

Built-in SNMPv2 MIB

Web RCS: highly intuitive HTML5-based user interface

Live video thumbnails shown on GUI

Fully configurable and operable from the front panel

Advanced audio processing and dynamic routing

Innovative 'Quick Preset' feature

Custom output formats for non-standard display applications

Area of Interest feature to customize active areas of outputs

Freeze of inputs, layers and screens

Cut and Fill at input level

Easily create and recall memories on your screen and AUX outputs

Fully functional simulator for offline configuration and practice

Quiet: 45dB average noise at 1m

Dedicated BNC with loop for Genlock, Blackburst and tri-level

Remote services and maintenance

Backup and restore functions

Technical specifications

10 INPUTS

4x HDMI 2.0

- ▶ up to 4K60 8-bit 4:4:4 or 4K60 12-bit 4:2:2

2x DisplayPort 1.2

- ▶ up to 4K60 10-bit 4:4:4 or 4K60 12-bit 4:2:2

2x 12G/6G/3G-SDI (level A & B)

- ▶ up to 4K60 10-bit 4:2:2

2x HDMI 1.4 or 3G-SDI (user selectable input plug)

- ▶ up to 2K60 10-bit 4:4:4 (HDMI) or 2K60 10-bit 4:2:2 (SDI)

Support custom input formats up to 4096 width / height

2 HDMI OUTPUTS + 2 MIRRORRED SDI OUTPUTS

2x HDMI 2.0 + 2x mirrored 12G/6G/3G-SDI (level A & B) connectors

- ▶ up to 4K60 8-bit 4:4:4 or 4K60 12-bit 4:2:2 (HDMI)

- ▶ up to 4K60 10-bit 4:2:2 (SDI)

Support custom output formats up to 4096 width / height

1 MULTIVIEWER OUTPUT

1x HDMI 1.4 + 1x mirrored 3G-SDI up to 2560x1600@60 8-bit 4:2:2

17 resizable widgets - customizable layouts with 20 memories

Monitor the Program, Preview, auxiliary screens and all the inputs

Built-in clocks, countdown and timers

LIVESTREAMING

RTMP/RTMPS livestreaming of inputs, outputs or Multiviewer

Compatible with various Web video platform (Youtube, Facebook, Twitch.tv...) , with personal credentials management

Compatible with local clients, such as OBS software

Video encoded in H.264 at up to 720p30, up to 6 Mbps

Embed any Audio stereo channels (AAC encoding)

PROCESSING

Based on Analog Way exclusive 5th generation scaling engine

Ultra low latency (16msec @60Hz)

BT.601; BT.709; BT.2020 color spaces support

Split layers mode: double the live layer count (no crossfade effect)

Advanced EDID management

Keying, Area of Interest and advanced output/screen test patterns

Built-in SDR/HDR converters

Colorspace conversion & color corrections based on 3D LUTs

Compatible with HDCP 1.4 and HDCP 2.2

TRANSITIONS & EFFECTS

Seamless transitions for mixing layers and foregrounds

Layer border effects and colors with separate shadow

Multiple transition effects

Layer effects: H/V Flip, Transparency, Crop and Mask

Cut and Fill at input level

Color effects: B&W, Negative, Sepia and Solarize

STILL IMAGES

4x 4K concurrent still images for the foreground layer

4x 4K concurrent still images for the background

Alpha-channel support

Still image library with 50 memories

Multi-file download/upload via Web RCS

Capture live content from any input or output

ADVANCED AUDIO PROCESSING

De-embed/embed audio channels on all inputs/outputs (PCM)

- ▶ 8 channels per HDMI 2.0, 12G-SDI or DisplayPort 1.2 input
- ▶ 4 channels per 3G-SDI input and 2 channels per HDMI 1.4 input
- ▶ 8 channels per active output

Level control, Mute, Balance, Delay, 1kHz Sinus Pattern features

1 VU meter display of a selected input on the multiviewer

Fixed or dynamic routing (breakaway or top layer)

Optional Analog and Dante™ audio networking card

- ▶ 2 analog stereo mini jack line in and line out 1/8" (3.5 mm)
- ▶ 32x32 Dante™ audio channels @48kHz
- ▶ Dual redundancy Ethernet ports - AES67 support

QUICK PRESET FUNCTION

3 Quick Preset modes : Fade to Black, Image or Master Preset

Audio routing configurable to follow the Quick Preset mode

CONTROL

Full-featured Front Panel with 480x272 color LCD screen

Web RCS: on-board intuitive web-based user interface

HTTPS for secure connection with downloadable certificate and key

Built-in SNMPv2 MIB

Shot Box² / Control Box³: Cost effective control solutions

RC400T: Ergonomic event controller

Compatible with SKAARHOJ Air fly Pro controller (AIR-FLY-PRO-N-V3B)

Compatible with Companion solution

Simple REST API (HTTP/S) and advanced JSON TCP protocol

Creston drivers & AW VideoCompositor (Premium Creston GUI)

OTHER FEATURES

Live monitoring of inputs & PGM on the large color TFT LCD display

200 user definable screen memories and 50 master memories

Dedicated BNC with loop for Genlock, Blackburst and tri-level sync

I/O connector status LEDs for easy troubleshooting

USB host type A: firmware update, configuration import/export...

Fully functional simulator for offline configuration & practice

Gigabit Ethernet via Neutrik RJ45

TSL protocols (v3.1 and 4.0)

Dimension (in Rack Units – RU)

- ▶ 2RU

Dimension (without rack ears & mounting)

- ▶ W 17.32" x H 3.46" x D 17.08"
- ▶ L 440 mm x H 88 mm x P 434 mm

Dimension (with handles)

- ▶ W 19" x H 3.46" x D 18.66"
- ▶ L 482.5 mm x H 88 mm x P 474 mm

Weight without accessories

- ▶ 7.60 kg / 15.43 lbs

Shipping weight (accessories included)

- ▶ 10.60 kg / 22.04 lbs

Operating conditions

- ▶ Temperature: 0 to 40°C (32 to 104°F)
- ▶ Humidity: 10% to 80%, non condensing

Noise (@1,6m height @25°C)

- ▶ Front: 45 dB@1m
- ▶ Rear: 45 dB@1m

Thermal dissipation

- ▶ 273 BTU/hr

Warranty

- ▶ 3-year warranty on parts and labor back to factory. Broken connectors are not covered by warranty

Power Supply

- ▶ 100-240 VAC, 3A 50/60Hz
- ▶ Fuse cartridge F4A 250VAC 5x20mm
- ▶ Max consumption: 95 W

EMC & Environnemental Compliance

- ▶ EN55032, EN55024, EN61000, FCC part15, ICES

Safety compliance

- ▶ IEC/UL/EN 62368-1
- ▶ CSA22.2#62368-1
- ▶ UL Listed (E359143)

Supplied with

- ▶ 1x Power supply cord
- ▶ 1x Remote Control Software (Web RCS)
- ▶ 1x Rackmount kit
- ▶ 1x Ethernet cross-cable
- ▶ 1x User Manual (pdf)
- ▶ 1x Quick Start Guide including safety instructions

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① Specifications subject to change without prior notice